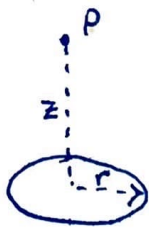


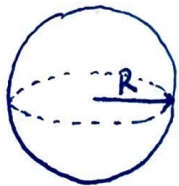
1) Answer the following questions about electrostatics:

a)



Find the electric field a distance z above the center of a circular loop of radius r , which carries a uniform line charge λ .

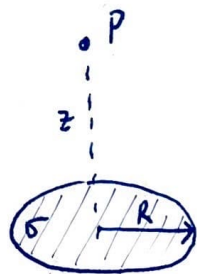
b)



i) Find the electric field outside a uniformly charged solid sphere of radius R and total charge q .

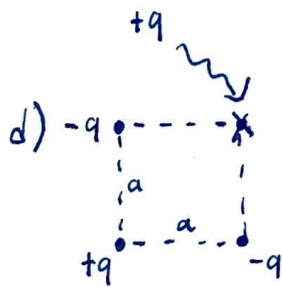
ii) If the charge density of the sphere is denoted by ρ , find the electric field inside the sphere.

c)



A uniform surface charge is given. Find the potential at a distance z .

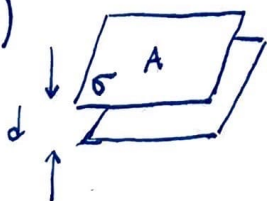
(Hint: • Consider a segment of the surface charge with radius r and circumference $2\pi r$. It behaves as a point charge if you consider the product $2\pi r dr \sigma$
• To calculate the integral make the substitution $u = z^2 + r^2$)



i) Three charges are situated at the corners of a square as shown. How much work does it take to bring in another charge, $+q$, from infinity (very far away) and place it in the fourth corner?

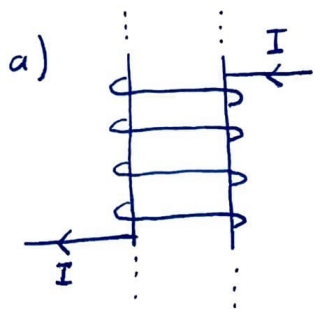
ii) How much work does it take to assemble the whole configuration of four charges?

e)



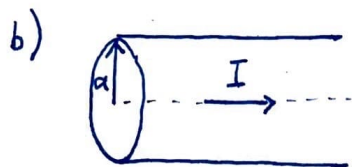
Find the capacitance of a parallel-plate capacitor shown in the figure.

2) Answer the following questions about magnetostatics:



Find the magnetic field inside a very long solenoid (ideally infinitely long), consisting of n closely wound turns per unit length on a cylinder of radius R and carrying a steady current I .

(Given: The magnetic field outside an infinite solenoid is zero)

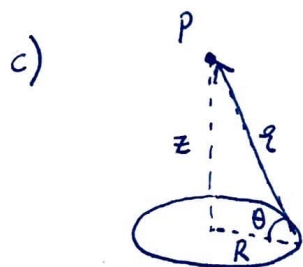


Suppose the current density in a wire of circular cross section, with radius a , is proportional to the distance from the axis:

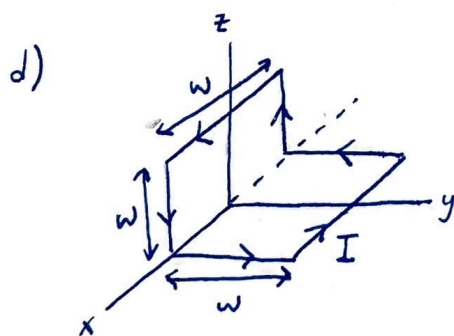
$$J = ks \quad (k \text{ is constant})$$

Find the total current in the wire.

(Given: Area element in polar coordinates is $da = s ds d\phi$)



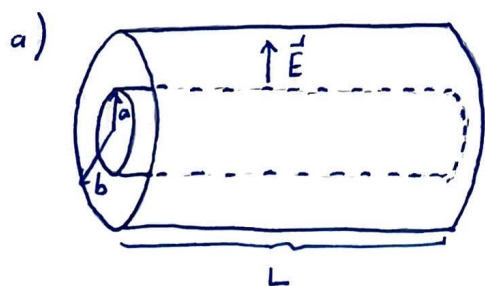
By using Biot-Savart law find the magnetic field a distance z above the center of a circular loop of radius R , which carries a steady current I .



Find the magnetic dipole moment of the loop shown in the figure. It doesn't look so, but all sides have length w , and it carries a current I .

e) An infinite solenoid (n turns per unit length, current I) is filled with linear material of susceptibility χ_m . Find the magnetic field inside the solenoid.

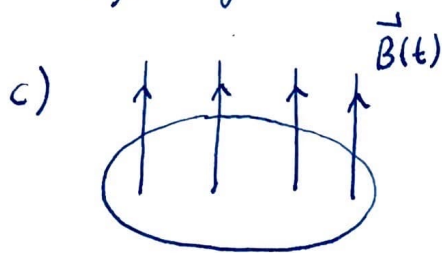
3) Answer the following questions about electrodynamics:



Two long cylinders (radii a and b) are separated by material of conductivity σ . If they are maintained at a potential difference V , what current flows from one to the other, in a length L ?

(Given: λ : charge per unit length on the inner cylinder)

b) Explain the flux law $\mathcal{E} = - \frac{d\Phi}{dt}$ briefly. What does that minus sign say us?



A uniform magnetic field, pointing ~~straight~~ straight up, fills the shaded circular region in the figure. If $\vec{B}$ is changing with time, what is the induced electric field?

(Hint: Leave $\frac{dB}{dt}$ as it is in your result)

d) Find the self-inductance of a toroidal coil with rectangular cross section (inner radius a , outer radius b , height h), which carries a total of N turns.

4) a) Write down Maxwell's equations in differential form and explain them very briefly.

b) By using Maxwell's equations derive the wave equation of an electromagnetic wave in regions of space where there is no charge or current.

(Hint: $\vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla} (\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}$)